



Green infrastructure and natural resource utilization for green development in selected belt and road initiative countries

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ARTICLE INFO

JEL classification:

A13

F64

C23

Keywords:

Green infrastructure

Sustainable development

Fossil fuels efficiency

BRI economies

Panel data approach

ABSTRACT

This paper measures the impacts of green infrastructure and fossil fuels consumption efficiency on the green development of 34 BRI (Belt and Road Initiative) economies from 2010 to 2021, measured by the CS-ARDL (Cross-sectionally augmented autoregressive distributed lag) panel estimation technique. The empirical results confirmed that the green infrastructure index only has a long-term positive effect on green development. The efficiency of exploitation of fossil fuel resources, both in the short term and in the long term, is a factor in advancing the goals of green development in the studied countries. Furthermore, there is an uni-directional causal association between the green infrastructure index and sustainable development index, and a bi-directional causality nexus exists between sustainable development and fossil fuels efficiency. The BRI economies should implement significant policies like developing green financial infrastructure, promoting reduction, reuse, recycle (the 3Rs) policies, and establishing a green economic development cooperation network.

1. Introduction

Sustainable development in terms of economic and social growth and progress in terms of environmental benefits is considered an essential and priority goal for the world's countries. Many scholars like Horton and Horton (2019) and Vuuren et al. (2022) believe that the future of human life is only possible by implementing sustainable development goals. Because due to the excessive depletion of natural resources, waste generation, climate change, and global warming, the conditions of human life in many geographical parts of the planet can be predicted shortly, and therefore the factor of sustainable development as a means of reducing and controlling environmental concerns should be more should be considered worldwide. Despite the international community's acceptance of the importance of sustainable development, in recent decades, the speed of progress of this discourse has not been acceptable, and the dependence of economic growth on fossil fuels and increasing carbon dioxide emissions are still on the border of threats. World Economic Forum (2022) reports that global carbon dioxide emissions reached a historical record of 36.6 billion tons in 2022, confirming the aggregate failure of sustainable development policies worldwide. In

addition, NASA (2023) expresses that the global warming threat continued in 2022 (the fifth warmest record since 1880) and was 0.89 °C higher than the baseline from 1951 to 1980. Due to the existence of environmental problems and their increasing trend in the future, in-depth practical study in sustainable development is now of great importance and necessity.

One of the primary components of environmental issues is the efficient use of fossil fuel resources. The natural resources that have existed on the planet for free for humans have been increasingly exploited since the British industrial revolution in the 18th century, and this unfavorable exploitation of natural resources has caused the emergence of threats such as disruption of the ecosystem and biodiversity (Hussain et al., 2020; Ali et al., 2021). Baste and Wantson (2022) mention that the rapid growth of natural resource utilization to satisfy industrialization has had various adverse environmental impacts. According to the Global Footprint Network (GFN) (2022), the rate of human consumption of natural resources is 75% more than the speed of ecosystem replenishment. However, efficient resource utilization can help decrease resource utilization and promote environmental protection. He et al. (2022) highlight the importance of the efficiency of resource deployment to

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neutralize environmental threats shortly. In other words, resource utilization means the management of available resources in the present to transfer them to posterity.

The presence of green infrastructure is another crucial requirement to achieve sustainable development. Green infrastructures are considered necessary to promote ecosystem services and green GDP flourishing. Wang and Banzhaf (2018) define GI (green infrastructure) as an approach comprising ecosystem and human well-being. According to Ye et al. (2022) and Sun et al. (2023a, c), green infrastructure has a multidimensional feature and tries to fill in economic, social, and technological gaps to reach sustainable development in a country. Green infrastructure is like a bridge that can bring the economy's green sector to sustainable development goals. The presence of reliable and advanced infrastructure will not allow the green economy to be converted into sustainable development goals.

The primary academic goal of this research is to evaluate the impacts of resource efficiency and green infrastructure on green development. We choose the BRI (Belt and Road Initiative) countries as the case study for this research. The following reasons have motivated us to study the BRI economies. First, among BRI, some major global carbon emitters (e.g., China, Indonesia, Russia) are trying to promote sustainable development and environmental protection policies. Wu et al. (2021) declare that the BRI economies almost have an environmental pollution problem and should be more careful about the natural resource depletion challenge.

Furthermore, the BRI countries have faced a depletion of natural resources, an element for economic growth. However, deploying natural resources affects biodiversity and the ecosystem (Hussain et al., 2020). In addition, China as the pioneer of the project of BRI, tries to promote infrastructure development in the members' geographical borders, which would align with environmental protection policies (Sun et al., 2022a, 2022c).

The main contributions of this research to earlier literature can be summarized as follows: First, the multi-dimensional green infrastructure index is calculated for the examined BRI countries. The index is novel, and we propose it for the first time in this paper. Second, the resource efficiency index is calculated based on Zhao and Rasoulinezhad's (2023) method for the examined BRI economies. Third, the sustainable development index (comprising education, life expectancy, marital footprint, carbon emissions, and income) is calculated based on Hickel (2020) for the examined countries. Fourth, the relationship between resource efficiency, green infrastructure, and sustainable development is estimated through the panel model of the BRI-examined countries (Sun et al., 2022b). The primary research limitation would be access to data on the variables of the BRI economies. With sufficient data, the analysis would be reliable.

The empirical results confirmed that the green infrastructure index only has a long-term positive effect on green development. The efficiency of exploitation of natural resources, both in the short term and in the long term, is a factor in advancing the goals of green development in the studied countries. Furthermore, there is an uni-directional causal association between the green infrastructure index to the sustainable development index, and a bi-directional causality nexus exists between sustainable development and resource efficiency. The BRI economies should implement significant policies like developing green financial infrastructure, promoting reduction, reuse, recycle (the 3Rs) policies, and establishing a green economic development cooperation network.

In continues, the paper comprises Section 2 (review of earlier literature), Section 3 (discussion on the theoretical baseline), Section 4 (expression of data description and estimation process), Section 5 (empirical findings and discussion), and the last Section of conclusions and policy implications.

2. Literature review

A strand of literature has focused on green recovery and sustainable

development in different countries. Capasso et al. (2019) reviewed over 113 earlier literature and concluded that green development is a chain of sustainable values, including skills, technology, and trade. Cuiyun and Chazhong (2020) explained that green development comprises all policies and instruments supporting environmental protection and resource conservation. Ruggerio (2021) highlighted the necessary conditions of a circular economy and green economy to promote policies toward sustainable development. Maiti (2022) studied the aspects of green growth in 32 economies from 2000 to 2015. Employing the panel threshold technique, they confirmed that sustainable growth is a process where markets and mechanisms have been transformed into eco-friendlier features. Lee et al. (2022) evaluated the green development-culture relationship in 122 economies. The results depicted that green development is more than just an economic issue. Indeed, it is influenced by various factors like social culture and the perception of people toward greening development. Chaaben et al. (2022) studied the sustainable growth in Saudi Arabia from 2015 to 2020. They concluded that sustainable growth is a complex concept associated with political, economic, social, and cultural variables. Ostergaard et al. (2022) expressed that sustainable development is all attempts to lower dependency on fossil fuels and to expand the deployment of renewable energy resources.

A second strand of literature studies the association between fossil fuel efficiency and green development. Zakari et al. (2022) tried to explore the nexus between eco-friendly development and energy resource efficiency in 20 Asia-Pacific economies from 2000 to 2018. The estimated coefficients by the Panel Correction Standard Error (PCSE) displayed that energy resource efficiency enhances energy transformation progress leading to less environmental pollution and natural resource depletion. Aldieri et al. (2022) explored the impact of resource efficiency on sustainable development in the USA, Japan, and the European Union from 2002 to 2017. The preliminary results confirmed resource utilization efficiency's direct and spillover effects on greening economic sectors and flourishing human well-being. Khan et al. (2023) studied the relationship between resource efficiency and environmental issues for four emerging Asian economies during 1990–2020. The results confirmed that resource efficiency could lead to technological progress, upgrading technologies, lowering carbon emissions, and promoting environmental protection. Sun et al. (2023b) addressed the resource efficiency impacts in E7 economies from 2010 to 2021. The results of the Generalized Method of Moments depicted the positive role of resource efficiency in increasing green GDP flourishing.

The third strand of literature comprises studies focusing on the nexus between green infrastructure and sustainable development. Chatzimentor et al. (2020) focused on the features of green infrastructure in Europe. They depicted that the green infrastructure is an eco-friendly network providing many sustainable services and leading to better living conditions on the planet of Earth. Chapman and Hall (2022) expressed that developing green infrastructure benefits the ecosystem and sustainable development through greening habitat connectivity and expanding green services. Miroshnyk et al. (2022) studied the nexus between green infrastructure and sustainable cities. They concluded that ecosystem services quality surges through enhancing sustainable infrastructure in the metropolis.

According to the review of earlier literature, there needs to be more literature about studying the relationship between resource efficiency, green infrastructure, and sustainable development in the case of BRI economies. To this end, the multi-dimensional green infrastructure index, resource efficiency index, and sustainable development index (comprising the dimensions of education, life expectancy, marital footprint, carbon emissions, and income) are calculated for the examined countries and, in continue the relationship between resource efficiency, green infrastructure, and sustainable development are estimated through the panel model of the BRI examined countries.

3. 3- Theoretical background

The impacts of green infrastructure and the efficiency of fossil fuel resources on green development can be interpreted from the point of some transmission channels. The efficiency of using natural resources can lead to the realization of green development through reducing resource consumption, increasing justice in access to resources, reducing energy poverty, creating balance in the earth's ecosystem, and reducing greenhouse gas emissions. Ullanwicz (2020) explained that resource efficiency leads to green development by balancing ecosystems and reducing disruption in biodiversity. Markolf et al. (2022) believed that resource efficiency could reduce the amount of energy resource exploitation leading to less carbon emissions and environmental destruction. However, resource efficiency needs technological progress. Advancement in innovative technologies is a factor in improving efficiency in the consumption of natural resources. In many economic sectors, such as electricity generation, transportation, and industrial production, the development of green technologies is an essential factor in reducing the consumption of fossil fuel resources, leading to the reduction of environmental pollution and a solution to the threat of climate change. On the other side, green infrastructure can provide many sustainable services to people in society. With the green infrastructure network, it will be possible to develop sustainable services for the beneficiaries of the green value chain, and therefore the sustainable development line will be interrupted. Having a suitable green infrastructure can lead to a reduction in the price of services in a country. In other words, a positive relationship exists between the development of green infrastructure and the reduction of inflation in an economy. Wang and Banzhaf (2018) and Yeo et al. (2022) declared that green infrastructure makes an efficient local network to ensure the quantity and quality of sustainable services. Under the existence of an efficient green infrastructure network, green development can be accessible in the country.

4. Data description and estimation process

In order to measure the association of green infrastructure, fossil fuels consumption efficiency with green development, 34 BRI countries (Albania, Algeria, Argentina, Armenia, Azerbaijan, Bahrain, Bangladesh, Belarus, Bolivia, Botswana, Bulgaria, Cambodia, Cameroon, Chile, China, Costa Rica, Croatia, Cuba, Cyprus, Czech Republic, Ecuador, Estonia, Fiji, Indonesia, Italy, Kazakhstan, Kuwait, Republic of Korea, Vietnam, Uzbekistan, Uruguay, United Arab Emirates, Turkey, and Singapore) are chosen to study from 2010 to 2021. The availability and accessibility of raw data of variables are two factors in selecting the examined countries. The sustainable development index (education, life expectancy, marital footprint, carbon emissions, and income) is calculated based on Hickel (2020) and added to the empirical model as the dependent variable. In addition, the multi-dimensional green infrastructure index (calculated based on the method of Heckert and Rosan (2016)) and the fossil fuels efficiency index (calculated based on Zhao and Rasoulinezhad's (2023) method) are considered explanatory variables. Consumer price index (CPI), patent, and FDI inwards are added to the model as the control variables. The preliminary information on the mentioned variables is reported in Table 1 as follows:

It is expected that the multidimensional index of green infrastructure has a positive effect on the sustainable development index. Because by improving the state of green infrastructure, the green value network spreads and grows in society and different economic sectors, which will lead to the realization of green development goals. It is also expected that the resource efficiency index coefficient will be positive. Because by increasing the efficiency of using fossil fuels resources, the process of fossil fuels resource destruction slows down, and also by reducing the use of resources in production and economic activities, the protection of the environment is improved. The consumer price index is expected to have a negative effect. Because the increase in the general level of prices

Table 1

The specifications of the selected variables.

Variable	Abbreviation	Unit	Source
Sustainable development index	SDI	–	Calculated based on Hickel (2020)
Multi-dimensional green infrastructure index	MGII	–	Calculate based on Heckert and Rosan (2016)
Fossil fuels efficiency index	REI	–	Calculated based on Zhao and Rasoulinezhad (2023)
Consumer price index	CPI	–	World Bank
Patent	PAT	number	The World Intellectual Property Organization (WIPO), World bank
FDI inward	FDII	Million US dollars	World Bank, Local websites

Source: Authors

will make the implementation of environmentally friendly projects more expensive, reduce the participation of the private sector in sustainable development projects and reduce the purchasing power of people in green markets. In the case of the patent variable, the vague effect can be positive or negative. Suppose the share of green patents in the total patents of a country is significant, with the increase in the number of these patents. In that case, the goals of green development are expected to be implemented. However, the share of green patents in total is insignificant. In that case, the possibility of adverse effects of the increase in patents on the green development of these countries will exist. There is such a forecast for the inflow of foreign direct investment. Suppose the flow of foreign direct investment into a country leads to the development of industries that consume natural resources. In that case, we expect negative coefficients of this variable in the empirical model. However, the share of green investment in the inflow of foreign direct investment is significant. In that case, we expect It is assumed that the sign of the foreign direct investment variable coefficient is positive.

The appropriateness panel estimator can be determined by some preliminary tests to estimate the coefficients of the independent variables. First, the cross-sectional dependency issue is tested by Pesaran (2015)'s CD technique (Equation (1)).

$$CD = \sqrt{\frac{2T}{N(N-1)}} \left(\sum_{i=1}^{n-1} \sum_{j=i+1}^n \theta_{ij}^t \right) \quad (1)$$

Next, the panel homogeneity test is run to find the panel framework's nature. If cross-sectional dependency among cross-section units is confirmed, the second generation of panel unit root tests should be employed. To this end, CIPS (cross-sectionally augmented IPS) and CADF (cross-sectionally augmented DF) are conducted. If the variables are stationary, then the panel co-integration test (Westerlund technique) is done to find whether the long-term association among variables exists. In order to estimate the coefficients, the cross-sectionally augmented autoregressive distributed lag (CS-ARDL) approach proposed by Chudik and Pesaran (2015) is conducted, which allows for cross-sectional dependency. Lastly, the FMOLS (Fully Modified OLS) and AMG (Augmented Mean Group) are employed as the robustness checks.

5. Empirical evidence and discussion

5.1. Estimation findings

First, it is necessary to check the existence of cross-sectional dependency in the cross-section units of the panel of the examined BRI economies. The results of the Pesaran CD test are listed in Table 2.

According to Table 2, it can be concluded that any unprecedented shock in a country leads to changes in the data of other countries, meaning that cross-sectional dependency exists in the panel of selected

Table 2
Pesaran CD findings.

Variable	Statistic	P-value	Abs (corr)
SDI	50.16	0.00	0.866
MGII	48.49	0.00	0.718
REI	35.01	0.00	0.660
CPI	41.19	0.00	0.719
PAT	39.88	0.00	0.583
FDII	0.524	0.00	0.972

Note: SDI, MGII, REI, CPI, PAT, and FDII stand for Sustainable development index, Multi-dimensional green infrastructure index, Resource efficiency index, Consumer price index, Patent, and FDI inward, respectively.

Source: Authors

countries. In continuation, the slope homogeneity in the selected series is explored and reported in Table 3 as follows:

The results, reported in Table 3, confirm that the panel of countries has the feature of heterogeneity. Next, the second generation of panel unit root tests - that consider the CD feature of the panel - are conducted. Table 4 represents the findings of CIPS and CADF panel unit root tests:

The findings of the CIPS and CADF tests reveal that all the variables are non-stationary at the level and become stationary at the first difference level. In other words, all the variables are $I(1)$, allowing the long-term association among variables to be checked. To this end, the Westerlund co-integration test adapted to CD presence in the panel is used. Table 5 reports the findings of the Westerlund panel co-integration test and depicts that our panel data are co-integrated and that there is a long-term relationship between variables.

Next, the cross-sectionally augmented autoregressive distributed lag (CS-ARDL) technique is conducted to estimate the independent variables' short-term and long-term coefficients.

According to Table 6, in the short term, the green infrastructure index does not significantly affect the green development of the countries of the One Belt One Road project. Some resource efficiency indicators have a positive and significant coefficient in the short term. In other words, with a 1% increase in the efficiency of exploitation of fossil fuel resources, the amount of green development in the studied countries will increase by 0.18%. Fossil fuels resource efficiency means less consumption of natural resources, protecting the environment, and expanding the fields of sustainable development. The consumer price index has a negative and significant coefficient in the short term. The interpretation is that inflation in the prices of goods and services is an obstacle to green development in the studied countries. Also, according to the results, patents and foreign direct investment have a negative and significant effect in the short term. With a 1% increase in the number of patents and the volume of foreign direct investment to selected countries in the One Belt One Road project, green development in these countries will decrease by 0.39% and 0.17%, respectively.

Estimating the coefficients of the independent variables in the long term shows that the green infrastructure and the efficiency of fossil fuel resources have positive and significant coefficients. In other words, in the long term, with a 1% improvement in green infrastructure and fossil fuel resource efficiency, sustainable development in selected One Belt One Road project countries will increase by 0.22% and 0.009%, respectively. Three variables of consumer price index, patent, and foreign direct investment significantly affect green development in the studied countries. With a 1% increase in the consumer price index, the number of patents, and the flow of foreign direct investment, the amount

Table 4
Second generation panel unit root tests results.

Variable	Cross-sectionally augmented IPS		Cross-sectionally augmented DF	
	At level	At first difference	At level	At first difference
SDI	-1.048	-6.184**	-1.849	-4.193**
MGII	-1.660	-5.366**	-2.117	-5.227**
REI	-1.045	-6.019**	-1.885	-4.100**
CPI	-0.989	-4.717**	-1.043	-5.994**
PAT	-1.115	-5.684**	-2.312	-9.449**
FDII	-0.786	-4.339**	-1.164	-7.129**

Note 1: SDI, MGII, REI, CPI, PAT, and FDII stand for Sustainable development index, Multi-dimensional green infrastructure index, Resource efficiency index, Consumer price index, Patent, and FDI inward, respectively.

Note 2: ** shows significance at 5% level.

Source: Authors

Table 5
Results of Westerlund panel co-integration test.

	Statistic	probability
G_t	-3.194	0.00
G_a	-10.847	0.00
P_t	-7.008	0.00
P_a	-6.146	0.00

Source: Authors

Table 6
The cross-sectionally ARDL estimation results.

Perspective	Independent variable	Coefficient	P-value
Short-term	Δ MGII	0.013	0.443
	Δ REI	0.182	0.003
	Δ CPI	-0.003	0.044
	Δ PAT	-0.391	0.019
	Δ FDII	-0.173	0.008
Long term	MGII	0.221	0.018
	REI	0.009	0.022
	CPI	-0.192	0.003
	PAT	-0.019	0.000
	FDII	-0.327	0.049
	ECM	-0.778	0.003

Note: MGII, REI, CPI, PAT, and FDII stand for Multi-dimensional green infrastructure index, Resource efficiency index, Consumer price index, Patent, and FDI inward, respectively.

Source: Authors

of sustainable development in selected countries of the One Belt One Road project decreased by 0.01%, 0.32%, and 0.77%, respectively.

Considering the short-term and long-term results together, the green infrastructure index only has a long-term positive effect on the green development of the One Belt One Road countries. This finding correlates with Wang and Banzhaf (2018) and Yeo et al. (2022), who confirmed the positive role of green infrastructure in the sustainable growth of countries. The efficiency of exploitation of fossil fuel resources, both in the short term and in the long term, is a factor in advancing the goals of green development in the studied countries. Meanwhile, the long-term effect of this variable on green development is more than its short-term effect. The finding of the positive coefficient of resource efficiency supports the earlier literature like Ullanwicz (2020), Markolf et al. (2022), Zakari et al. (2022), Aldieri et al. (2022), and Khan et al. (2023). Inflation in both the short and long term is an obstacle to realizing green development in the countries of the Belt and Road project. The increase in prices causes a decrease in the production power of companies, the purchasing power of households, and as a result, the inability of the leading players in the economic markets to interact and support greening economic policies. This result is in line with Dabbous and Tarhini (2021), who emphasized the negative impact of inflation on

Table 3
Panel slope homogeneity test.

-	Value	Prob.
Δ	30.154	0.00
adj	35.448	0.00

Source: Authors

sustainable economic development. The two variables of patents and foreign direct investment flow hurt the green economic development of the studied countries in the short and long term. The interpretation of the negative coefficients of these two variables is that the share of green innovation (green patents) and green investment in total patents and foreign direct investment is low. Therefore any increase in patents and foreign direct investment causes more depletion of fossil fuels resources, emission of carbon dioxide, and waste generation.

In order to ascertain the validity of empirical findings, two other estimators of FMOLS and AMG are employed to re-estimate the long-term coefficients. Table 7 lists the results of the robustness check:

Lastly, the Dumitrescu-Hurlin panel causality test is carried out to explore the direction of causality associations. Table 8 represents the results of panel causality relationships.

The causality test results confirm an uni-directional causal association between green infrastructure and sustainable development indexes. In contrast, a bi-directional causality nexus exists between sustainable development and resource efficiency (confirmation of the feedback hypothesis).

5.2. Discussion: does income level matter?

The results obtained and interpreted in the previous section are about the set of selected countries in the One Belt One Road project, which includes countries with a high-income level and countries with a low and middle-income level. A critical discussion is that the income level filter may be necessary, and the relationship between dependent and independent variables is different in countries with a high-income level and countries with a low and middle-income level. The critical role of the income level of countries in the association with variables has been confirmed by a group of earlier studies like Rasoulinezhad and Mostaghimi Ghomi (2022) and Taghizadeh-Hesary et al. (2021). The reason for the difference in the relationship between the variables in high and low-income countries is that high-income countries usually have the better financial infrastructure, more economic resilience, more capital accumulation, less risky business environment, and more developed physical and human capital in sustainable development. In order to test the importance of income level in the current research problem, eight high- and upper-middle-income countries (Bahrain, China, Italy, Kazakhstan, Republic of Korea, United Arab Emirates, Turkey, and Singapore) are selected from the sample of countries of the One Belt One Road project and the coefficients are estimated separately.

The coefficient estimation results reported in Table 9 confirm that income level plays a vital role in influencing the variables on the green development of the One Belt One Road countries. The index of green infrastructure in countries with a high and upper-middle income level in

Table 7
Robustness check.

Estimator	Independent variable	Coefficient	P-value
FMOLS	MGII	0.143	0.002
	REI	0.029	0.000
	CPI	−0.158	0.019
	PAT	−0.029	0.008
	FDII	−0.023	0.034
AMG	MGII	0.685	0.011
	REI	0.448	0.039
	CPI	−0.333	0.005
	PAT	−0.119	0.054
	FDII	−0.078	0.000

Note: MGII, REI, CPI, PAT, and FDII stand for Multi-dimensional green infrastructure index, Resource efficiency index, Consumer price index, Patent, and FDI inward, respectively.

The findings estimated by the FMOLS and AMG support the signs of empirical coefficients reported in Table 6, confirming the model's reliability and conclusions.

Source: Authors

Table 8
Dumitrescu-Hurlin panel causality test.

Ho	W statistic	P-value
SD→MGII	2.948	0.668
MGII→ SD	4.111	0.009
SD→REI	1.103	0.031
REI→ SD	2.889	0.044

Note: SDI, MGII, and REI stand for Sustainable development index, Multi-dimensional green infrastructure index, and resource efficiency index, respectively.

Source: Authors

Table 9
CS-ARDL estimation for the high and upper-middle income group.

Perspective	Independent variable	Coefficient	P-value
Short-term	ΔMGII	0.129	0.003
	ΔREI	0.304	0.019
	ΔCPI	−0.033	0.003
	ΔPAT	0.019	0.018
	ΔFDII	0.059	0.042
Long term	MGII	0.495	0.053
	REI	0.329	0.000
	CPI	−0.119	0.392
	PAT	0.200	0.003
	FDII	0.192	0.048
	ECM	−0.818	0.000

Note: MGII, REI, CPI, PAT, and FDII stand for Multi-dimensional green infrastructure index, Resource efficiency index, Consumer price index, Patent, and FDI inward, respectively.

Source: Authors

the One Belt One Road project has a positive and significant coefficient both in the short and long term. Also, the consumer price index has an adverse effect only in the short term, and in the long term, the effect of this variable on green development is not significant. Moreover, the sign of the coefficients of two variables, patent, and foreign direct investment, is positive and significant, expressing that with the increase in the number of patents and foreign direct investment, the amount of green economic development index in high and upper-middle-income countries in the BRI project will improve.

6. Conclusions and policy recommendations

Sustainable development is considered a priority goal for countries worldwide. Many scholars believe that the future of human life is only possible by implementing sustainable development goals. This paper evaluated the impacts of resource deployment efficiency and green infrastructure on the development of the 34 BRI economies from 2010 to 2021 through the panel data approach. According to the empirical findings, the green infrastructure index only has a long-term positive effect on the green development of the One Belt One Road countries. The efficiency of exploitation of fossil fuel resources, both in the short term and in the long term, is a factor in advancing the goals of green development in the studied countries. Inflation in both the short and long term is an obstacle to realizing green development in the countries of the Belt and Road project. The two variables of patents and foreign direct investment flow hurt the green economic development of the studied countries in the short and long term.

Regarding the causality associations, the concluding remarks are the uni-directional causal association from the green infrastructure index to the sustainable development index, and a bi-directional causality nexus exists between sustainable development and resource efficiency. In addition, the income level is essential in evaluating the relationship between variables. In the case of high and upper-middle-income BRI economies, the index of green infrastructure has a positive and significant coefficient both in the short and long term. Also, the consumer price

index has an adverse effect only in the short term, and in the long term, the effect of this variable on green development is not significant. Also, the sign of the coefficients of two variables, patent, and foreign direct investment, is positive and significant.

Regarding the policy implications, the countries of the One Belt, One Road project should pay more attention to green financial infrastructure. Because through the financial green infrastructure network, it is possible to develop green investment and advance the goals of green technology progress. Sustainable financial infrastructure will enable the increase of private sector participation, the transparency of green markets, and the implementation of environmentally friendly projects in the countries of the One Belt One Road project. Implementing policies to increase the efficiency of the exploitation of fossil fuels resources, such as the policy of reduce, reuse, and recycle (the 3Rs), should be among the priorities of the countries' One Belt One Road project programs. It is proposed to establish a green economic development cooperation network in the countries of the One Belt, One Road project as an effective practical policy. In the countries of this project, some have more capital and knowledge that can provide financial and scientific assistance to other countries in this project of green development; fossil fuels efficiency and sustainable infrastructure.

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Despite the limitations of the research, like information availability and the lack of new official data, according to the authors of this research, it made significant contributions to the green development literature of the countries of the One Belt One Road project. However, in the future, the effect of the spread of the corona disease on green development, sustainable infrastructure, and the efficiency of fossil fuels resource exploitation in the BRI economies will be studied. Also, studying the country level and comparing the results between different countries in the BRI project can provide more accurate and practical results to policymakers.

Author statement

Bo Wang: Conceptualizing, writing, drafting-Original draft, and empirical estimations. **Haoting Yang:** Drafting-Original draft, and empirical estimations. **Chunyu Bi:** Conceptualizing, writing, drafting-Original draft. **Yanchao Feng:** Empirical estimations and Supervision.

Declaration of competing interest

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all the authors. We declare that there is no conflict of interest.

Data availability

Data will be made available on request.

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